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A Plagiarism Detector in Learning Management System for UTHM

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Abstract

This project was initiated to develop a plagiarism detector in a web-based learning management. Existing learning management systems often suffer from fragmented course management, limited real-time interaction, and insufficient data security, which reduces their effectiveness in supporting teaching, learning, and academic monitoring. The project was developed using web technology including Flask, HTML, CSS, JavaScript, and MySQL, based upon standard requirements analysis, system design and development, implementation, and testing for software systems. The results of the project indicate that the system allows for secure user authentication, structured management of courses, the ability to submit assignments on time, to monitor and analyse student performance through the use of analytics and to have a built-in plagiarsim detector. Therefore, in summary, it can be concluded from the findings of this study that the proposed web-based learning management system is an improved way of managing and enhancing academic operations.

1. Introduction

The purpose of this project is to provide Learning Management System (LMS) [1] at UTHM with the integrated plagiarism detection mechanism. It has been developed with the ultimate goal of reducing work load from lecturers as the similarities are reported based on automated comparisons between a database and an assignment instead of third-party tools. For freedom and openness, the architecture integrates blockchain and encrypted audit trail, secure authentication system in order to guarantee that academic records and plagiarism scores are never tampered with.

Although LMS platforms are widely used, they do not typically provide out-of- the-box support for ensuring data integrity, authenticity, and confidentiality. The lack of plagiarism tool does not encourage UTHM to ensure academic integrity and data protection effectively. And most of the current platforms are facing complex user interfaces, no real-time analytics is provided and lacks security such as logging activities to observe that there should be no unauthorized access and breaches for sensitive academic data.

The main aim of this project is building secured LMS with an auto plagiarism checker and designing advanced platform dedicated for UTHM users with performance indicators, real-time updates etc. It also provides the opportunity to obtained solid users feedback about our system through comprehensive functional and user acceptance testing.

The main outcome of this initiative is to create an integrated LMS for UTHM that will be a fully functional, data-driven, secure grade calculation, and will assist in the evaluation of submitted work for plagiarism. The integration of these features into a single LMS will enable lecturers to streamline their grading, enhance the time frame for providing feedback to students, and also protect sensitive intellectual property via superior encryption and authentication protocols.

This project is critical in strengthening UTHM's digital infrastructure by improving the quality of academic integrity and increasing operational efficiency. The incorporation of a plagiarism checking function in the LMS will not only reduce the cost of purchasing third party applications but also enhance the privacy of individualized data and foster an environment of openness and innovation within academia. Overall, the LMS will deliver a reliable, user-friendly product that responds to the requirements of higher education.

2. Related Work

The literature review begins by defining a Learning Management System (LMS) as a computer-based platform designed to centralize educational materials, assignment submissions, and communication between stakeholders.

2.1Manual or Traditional Study Assessment

The traditional or manual study assessment process utilizes the physical handling of printed documents for students' assignments. Therefore, the individual method of assessment requires a lecturer to manually review, comment, and record students' grades (which appear on hard-copy assignments). One of the major disadvantages of using a manual process is the difficulty a lecturer may have in determining whether a student has plagiarized from another source; in many cases, that determination will rely solely on the lecturer's memory and ability to observe. As a result of these issues with traditional manual methods of assessing student learning, many educators are now utilizing Electronic Assessment Systems as their means of assessment.

2.2 Learning Management System

An LMS is a centralized digital infrastructure to support the entire teaching-learning process for both students and lecturers. Lecturers can use the LMS to store all their teaching materials, assign and track assignments for their students, and record the level of engagement from each of their students with their teaching materials. Students can view all the materials, submit assignments, and keep track of their academic performance in one place. The implementation of an LMS has many practical advantages for students, the most important being the convenience of being able to learn on their own time and from any location in the world. Additionally, an LMS can also improve educational efficiency by automating many of the administrative responsibilities involved with grading and distributing course content, as well as allowing for enhanced communication through integrated messaging and notification systems. An LMS also allows for secure digital records of every academic interaction between students and their lecturers; it prevents data loss, is a transparent system of record, and supports a highly scalable digital learning environment [1].

2.3 Plagiarism

Plagiarism refers to an individual's use of another individual’s intellectual creation or writing without the permission of the original creator for the purpose of obtaining some type of unearned reward for this, which can occur intentionally or unintentionally. This act constitutes a serious violation of the principle of academic integrity in institutions of higher learning, as it diminishes the student's credibility and prevents them from truly progressing in their studies[2]. In order to support equitable treatment of all students and in support of a high level of ethical behavior, all universities have specific policies and procedures that monitor and ensure that all submissions made by students are entirely unique in their original work product.

Universiti Tun Hussein Onn Malaysia (UTHM) requires that all students maintain an upper 30% similarity index in their submissions (excluding bibliography and cited work). This level of the similarity index takes into consideration the naturally occurring similarities inherent in academic writing while enabling the work of the student to primarily represent their own comprehension of the material. By enforcing a maximum similarity index established at UTHM, the university reinforces ethical writing and allows students to be evaluated fairly on their own academic success [3].

2.4 Method or Technologies to Detect Plagiarism

With respect to policies regarding academic integrity, the proposed system has been developed in order to provide a structured approach to detect any instances of plagiarism in submitted student assignments. The lecturer creates an assignment in the LMS and once all student submissions are received, the LMS will automatically time-stamp each file and place all files in a queue. No comparison of files will occur until the deadline for final submissions has been reached. This approach allows for equitable treatment of all students submitting their assignments and does not give an advantage to those who submit their assignment before the deadline [4].

The analysis of the data is done with the help of three methods of technology:

Cosine Similarity[5]: This mathematical formula converts textual content into numerical vectors. To find how similar the two vectors are, this system computes the cosine of the angle created between both vectors. This enables the system to gauge similarity irrespective of the difference in size between the two pieces of content.

Term Frequency - Inverse Document Frequency (TF-IDF)[6]: This statistical method looks at both the frequency that a term appears in a document (TF), as well as how many times that same term appears across all documents considered for similarity comparison (IDF). By removing the significance of common/generic terms from all of the documents, the system can more easily and quickly identify if any of the other pieces of content have copied from or used any unique content.

N-grams: The N-gram technique is used to locate the occurrence of "chunks" of text (such as three consecutive words) between two pieces of content. Unlike a traditional word match comparison that relies on identifying individual words, the N-gram method compares phrases in a specific order. By calculating how similar phrases match each other, N-grams provide additional verification regarding improper use of content.

2.5 Study of Existing Related System

UTHM's AUTHOR LMS [7] doesn't contain a built-in plagiarism detection system so lecturers must use another service (e.g., Turnitin) to confirm originality. This creates inconsistency issues and adds additional time to when faculty must complete these tasks. Other notable online learning environments (i.e., Google Classroom[8] and Edmodo[9]) have experienced similar shortcomings with their plagiarism detection features. While Google Classroom is highly accessible and integrates with Google Workspace, many teachers need third-party subscriptions to access plagiarism detection options through Google Classroom. Likewise, Edmodo offers excellent opportunities for lecturers to enhance student collaboration and interaction but lacks an automated way to check for originality within students' assignments; as such, lecturers have to manually search for original submissions.

2.6 Comparison with Existing System

Table 1: Comparison Table

Features / System	AUTHOR	Google Classroom	Edmodo	Proposed System
Login	✓	✓	✓	✓
User Role Management	✓	✓	✓	✓
Upload / Submit Documents	✓	✓	✓	✓
Communication / Announcements	✓	✓	✓	✓
Assessment / Tracking Progress	✓	✓	✓	✓
Report & Analytics	✓	Limited	Limited	✓
File Sharing	✓	✓	✓	✓

Feedback System	✓	✓	✓	✓
Data Backup	X	✓	✓	✓
Security / Access Control	✓	✓	✓	✓
Built-in Plagiarism Detector	X	X	X	✓

In table 1, when comparing AUTHOR, Google Classroom, and Edmodo, it is clear that all three share the same major issue: none of them include built-in tools for detecting plagiarism. Both AUTHOR and Google Classroom are safe, reputable environments managed by universities, but they both rely on third-party services to check for originality after a student submits a paper. Edmodo focuses on improving student engagement and interaction; however, it does not offer advanced security options or similarity analysis. The new system solves these problems by adding an automatic plagiarism detection module into the student upload workflow. Each submission will now be scanned, analyzed, and securely stored to significantly decrease the likelihood of student cheating. By combining these features into one process, the system provides a consistent and reliable method for preserving the integrity of student's academic work and the university's online environment.

3. Methodology

Chapter Three describes how the project will use the Agile Model Methodology[10] to build the system through several iterations on a continual basis. The Agile Model Methodology relies heavily on team collaboration for refinement and enhancement of each module, based on ongoing feedback from lecturers and students. The methods employed to follow the Agile Model Methodology will consist of a structured workflow broken down into five phases; planning, analysis, design, development, and testing. By following this cycle of iterative design throughout development, a working prototype will eventually emerge that meets each stakeholder's (both students and lecturers) needs, and they will have a fully functionally compliant system with academic integrity compliance.

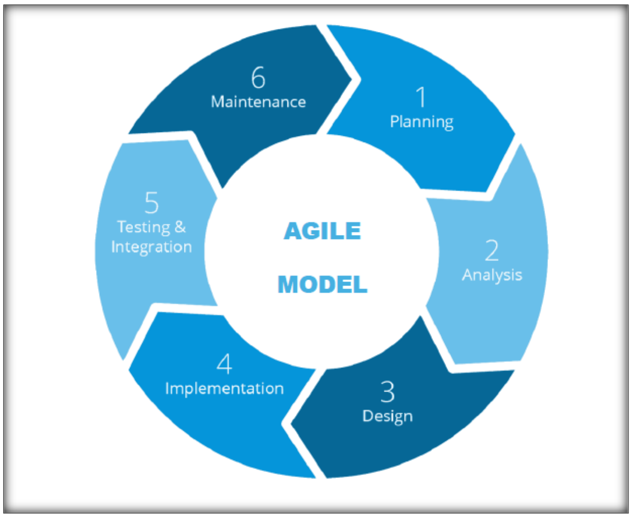

Fig. 1 Agile Model

During the Planning Phase, the goals and requirements for this project were established by having discussions with UTHM students and lecturers as well as the virtual learning portal. The project scope and timeline were established based on stakeholder discussions as well as the core features (uploading documents and similarity checking) of the system. As a result of these discussions, the user stories that will guide the development of the system were created.

The next phase is the Analysis Phase. This phase focuses on identifying the specific user needs in relation to how easy the process will be for students to upload their assignments and how automatically generated similarity reports can be produced for lecturers. This phase is also focused on defining all of the technical requirements needed to develop the system. The finalisation of algorithm selection, server load management and the workflow for storing data in relation to the limitations of the AUTHOR system were also performed.

In the Design Phase, the architecture of the system was established to ensure the ease and usability of the system by end-users. The key deliverables from this phase are: ERD's for the database structure, DFD's representing the process logic for the uploading of documents and similarity checking processes, and UI layouts designed using Figma, which will allow for stakeholder feedback before the coding of the system commences.

In the implementation phase, we will develop the system using a sprint-based incremental model that builds on four main functions of user login, document upload, similarity comparison and report generation. The back-end of the application will be developed using PHP, Python and MySQL, while the front-end of the application will use HTML, CSS and JavaScript. Each iteration of the application will be continuously enhanced with new features and improvements to the algorithms based on feedback received during the review process at the end of each sprint cycle.

The testing phase is responsible for testing the system to ensure its reliability through a three-part verification process: Unit Testing (testing of individual system modules), System Integration Testing (testing the transfer of data between modules without errors) and User Acceptance Testing (actual use of the system by lecturers and students to validate the usability of the system)[11].

Finally, the maintenance phase will be responsible for ensuring long-term viability and viability of the system by fixing bugs that have been reported, modifying detection algorithms to keep up with changes in academic standards, and adding security features or changes based on ongoing user feedback.

4. Analysis and Design

The chapter describes how academic honesty can be secured through the use of blockchain in learning management systems (LMS). First, this is done through detailed reviews of existing academic processes to find their limitations and to help create data-driven solutions. Next, as a way to keep track of all requirements needed for creating the system in an easy-to-follow manner, the requirements are expressed in various simplified format or representations using graphical representations. These graphical representations include CTD, DFD, and ERD. Finally, this chapter describes how to plan the database schema as well as how to design the user interfaces to provide the best possible experience for the various parties involved.

4.1System Requirements

Table 2 shows all the six modules in the system.

Table 2: System Module

No	MODULE	DESCRIPTION
1	Login Module	Ensures safe access for administrators, lecturers, and students via role-based access control, two-factor authentication, and login credentials.
2	Course Management Module	Enables creation, editing, and organization of courses, assignments, lecture materials, and announcements.
3	Assignment Submission & Plagiarism Module	Records submission history, offers real-time originality reports, automatically checks for plagiarism, and enables students to turn in work safely.
4	Notification & Reminder Module	Sends emails and pop-up notifications in real time about impending deadlines for assignments, plagiarism alerts, and course announcements.
5	Performance Analytics Module	Creates class-level statistics, plagiarism trends, submission patterns, and student performance dashboards to help lecturers make data-driven decisions.
6	Data Security Module	To avoid data loss or unwanted access, it makes sure that all system data, submissions, and reports are secured, safely stored, and routinely backed up.

4.1.1Functional and Non-Functional Requirements

Table 3 shows the list of functional requirements, Table 4 shows the categories of non-functional requirements.

Table 3: Functional Requirements

No	Module	Requirements
1	Login Module	• Authenticate users using secure credentials (username and password).
2	Course Management Module	• Allow lecturers to manage academic courses. • Allow students to view enrolled courses and relevant course information.
3	Assignment Submission & Plagiarism Module	• Enable students to upload assignment files (e.g., PDF, DOCX) for individual/group activities. • Generate a detailed similarity report comparing student work.
4	Performance Analytics Module	• Generate visual charts showing a student's similarity score trends over time.
5	Notification & Reminder Module	• Automatically trigger "nearing deadline" alerts to students 24–48 hours before an assignment is due. • Notify students immediately when a assignment is released.
6	Data Security Module	• Record all critical system transactions (lecturer/student activities) onto a local blockchain ledger[12].

Table 4: Non-Functional Requirements

NO	Requirements	Description
1	Security	• All sensitive data, including report hashes and student grades, must be stored in a blockchain to ensure immutability and prevent tampering.
2	Performance	• The plagiarism analysis and report generation should be completed within 10 seconds for standard document sizes.
3	Reliability	• The local blockchain ledger must be automatically verified for consistency every time a new block is added. • The system should provide a clear error message and log the incident if the database or blockchain connection fails.
4	Usability	• The user interface must follow a consistent colour scheme (orange and white for lecturer, blue and white for student) to align with institutional branding and improve user navigation.
5	Scalability	• The database schema must be designed to accommodate a growing number of courses and student enrollments without requiring structural changes.

4.1.2 User Requirement Analysis

Table 5 shows the user requirement table.

Table 5: User Requirement

No	Requirement Description
1	The lecturer accesses a secure dashboard to manage course-specific data and student records.
2	The lecturer broadcasts announcements and academic updates to the course stream for student viewing.
3	The lecturer configures assignment activities by defining titles, descriptions, and submission deadlines.
4	The lecturer evaluates student work through detailed plagiarism reports generated.
5	The lecturer validates the authenticity of academic reports via a blockchain-linked audit trail interface.
6	The lecturer monitors overall class performance and integrity trends through an analytics dashboard.
7	The student view enrolled course and announcements in the main dashboard.
8	The student uploads digital assignment files to the system before the designated deadline.
9	The student receives automated reminders regarding upcoming deadlines to prevent late submissions.

10	The student tracks individual academic progress through personalized performance analytics.
11	The student views notifications triggered by new assignments posted or near due assignments.

4.2 Context Diagram

A context diagram is a diagram that represents the entire system. The purpose of this diagram is to show the expected inputs and outputs from the system. This context diagram was created based on the user of the system perspective. Figure 2 shows the context diagram of Auto Plagiarism Check in LMS.

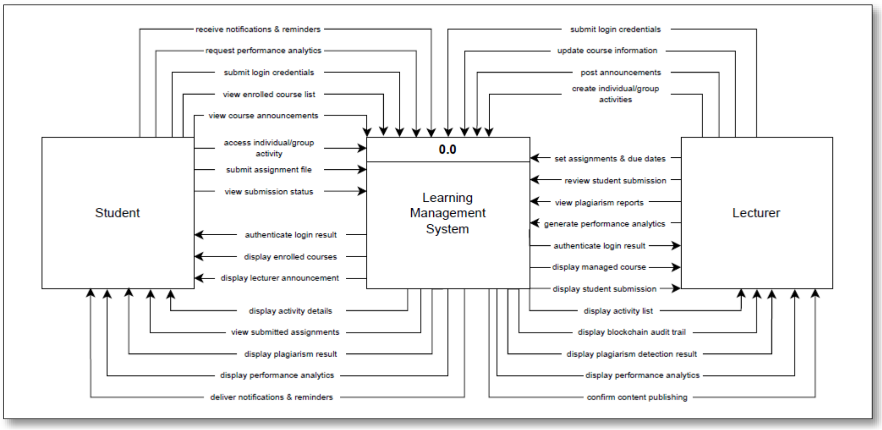

Fig. 2 Context Diagram

4.3 Data Flow Diagram Level 1 Process 3

Figure 3 shows the lecturers create assignment tasks and it will in the system's database. Students upload their completed files, which are automatically checked for plagiarism. The system then generates a report for the lecturer to assess the integrity and originality of each student assignment.

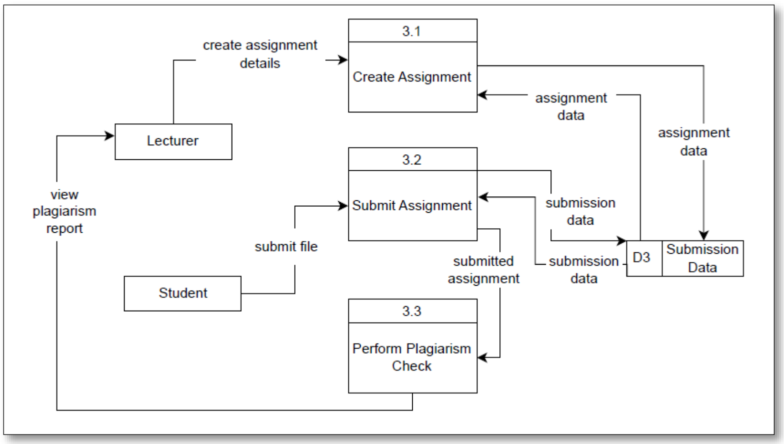

Fig. 3 DFD Level 1 process 3

4.4 Flowchart

Flowcharts use standardized symbols and directional arrows to convey a process or algorithm step by step. This visual representation provides a clear reference for both Development and Performance Stakeholders to assess the logical structure of a system, determine its decision-making points, and evaluate how it works. Figure 4 and Figure 5 shows the workflow of Auto Plagiarism Check in LMS.

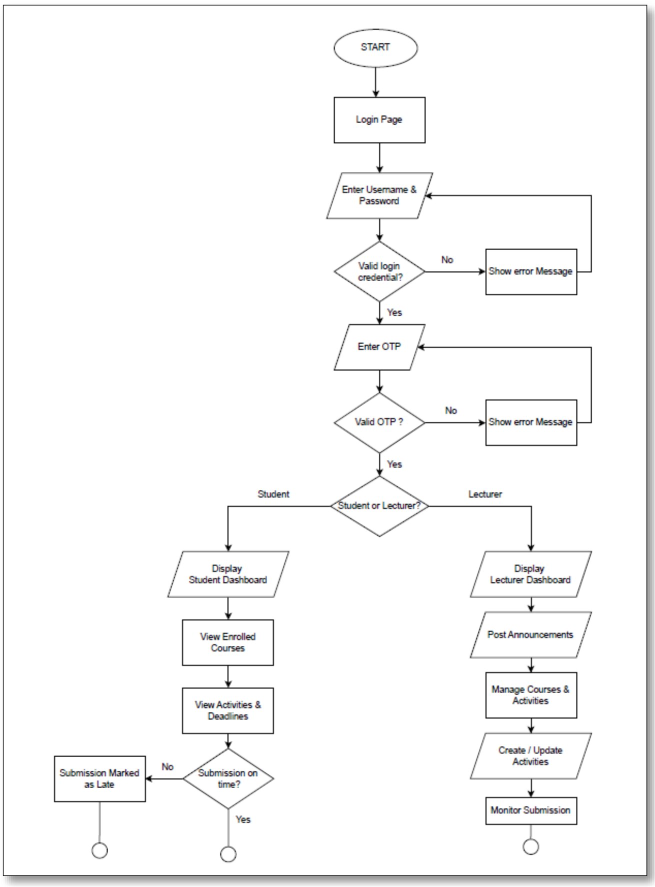

Fig. 4 Flowchart i

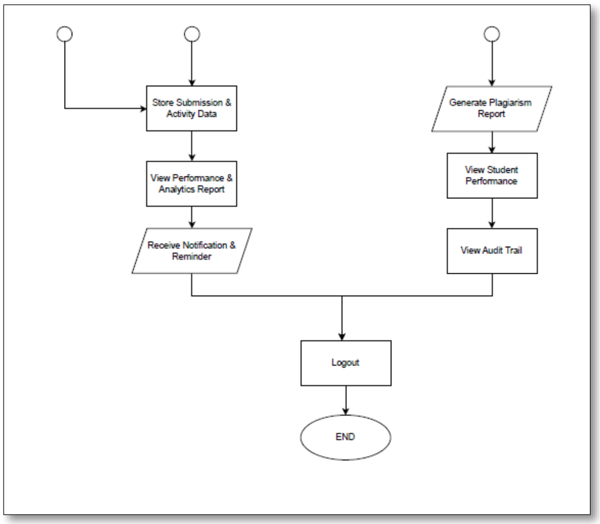

Fig. 5 Flowchart ii

4.5 System Architecture

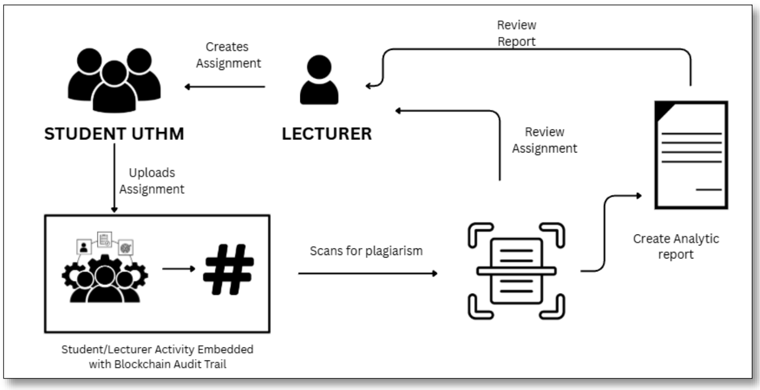

Fig. 6 System Architecture

Figure 6 shows that every time an assignment is submitted, an automated plagiarism scan will be performed. The results of the scan will generate a report containing an analysis of the quality and the originality of the student's work. This instantaneous and data-driven feedback to lecturers and students enhances learning efficiency by giving them the ability to provide real-time feedback regarding assignments and allows lecturers to assess assignments quickly.

4.6 Entity Relationship Diagram (ERD)

Entity Relationship Diagram (ERD) is a type of structural diagram for use in database design. Figure 7 below shows the ERD for UTHM Student Attendance System Using Blockchain.

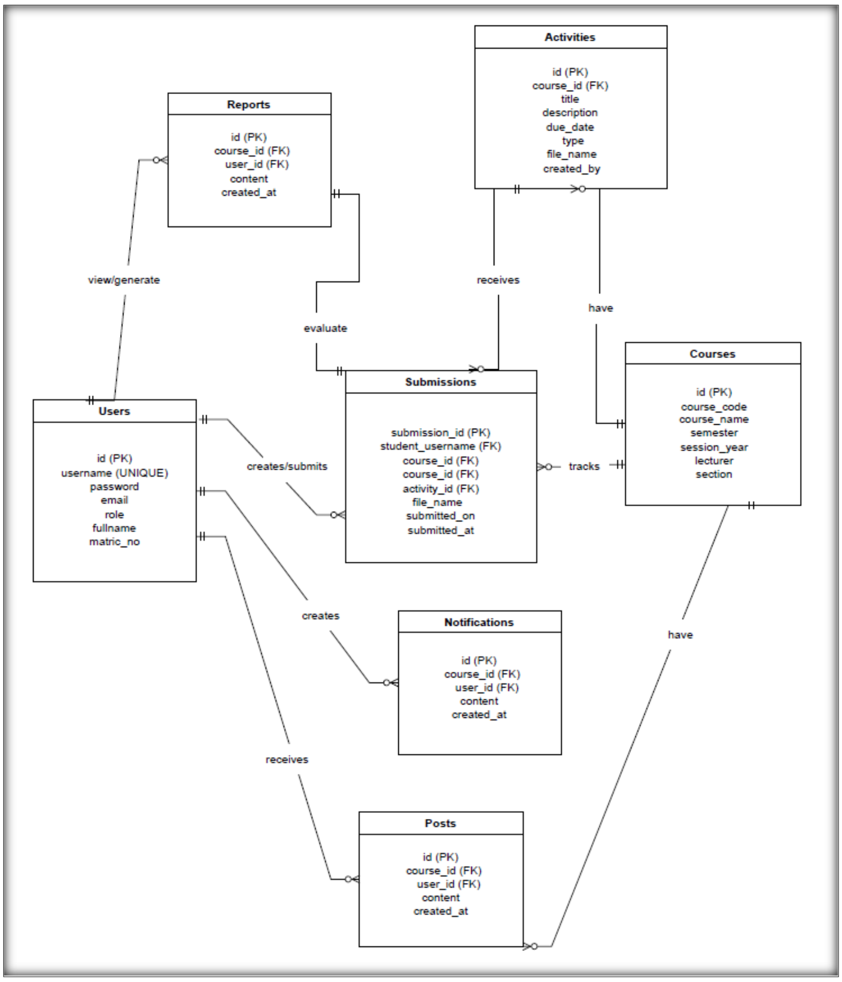

Fig. 7 ER Diagram

4.7 Interface Design

The clean, intuitive, and user-friendly design of the system interface ensures that students and lecturers can find their way to the features they need quickly and with the least amount of time spent learning how to use the system. Figure 8, Figure 9, Figure 10, Figure 11 and Figure 12 shows some interface design using figma[13] for this system.

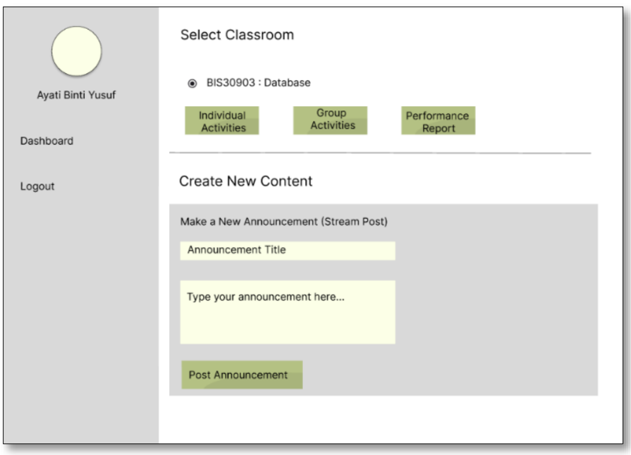

Fig. 8 Lecturer Dashboard

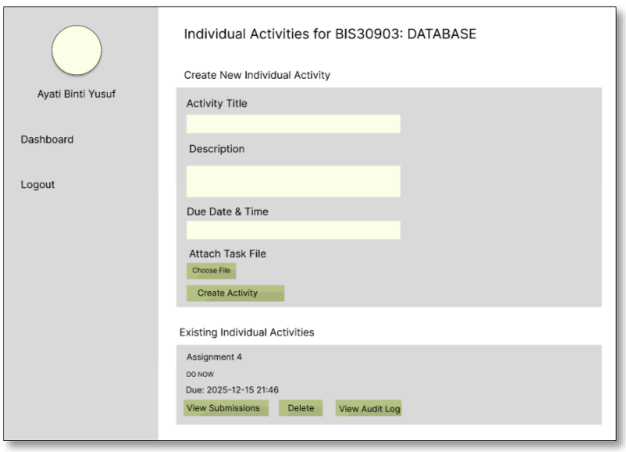

Fig. 9 Create Assignment

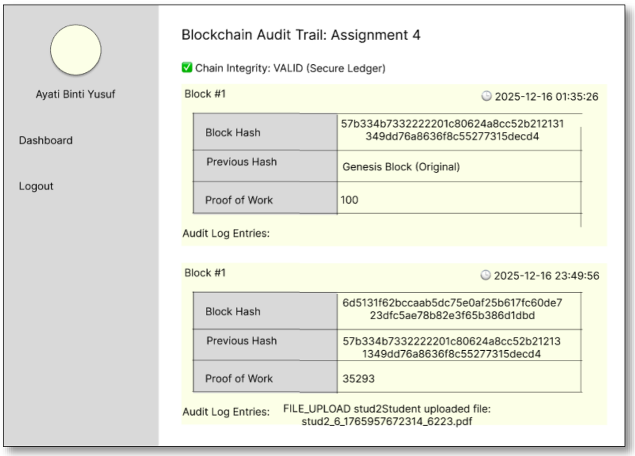

Fig. 10 Audit Trail

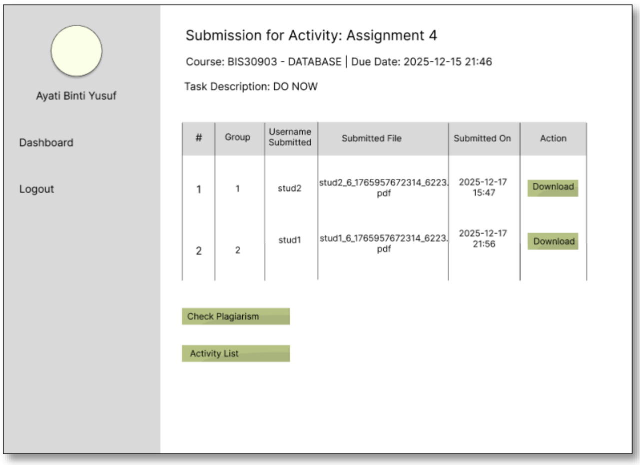

Fig. 11 Assignment Submission

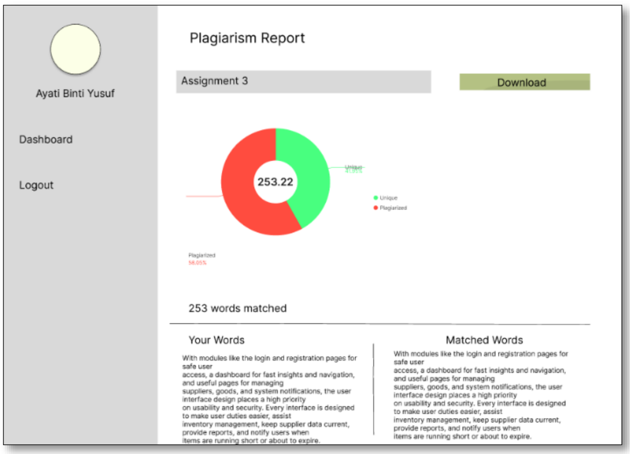

Fig. 12 Plagiarism Report

5. Conclusion

This project provides a secure, user-friendly learning management system that provides efficient management of courses, submissions for assignments and engaging students and lecturers together in academic interactions. With performance information available at the user's fingertips and notifications for keeping users updated and tracking their academic progress, users are always informed about their course activities. Furthermore, effective data security measures protect the integrity of both user and academic data within the system. Overall the Learning Management System improves the quality of teaching/learning by providing essential functionality along with compliance to security requirements typical of today's educational platforms.

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Conflict of Interest

Authors declare that there is no conflict of interests regarding the publication of the paper.

Author Contribution

This journal requires that all authors take public res
ponsibility for the content of the work submitted for review. The contributions of all authors must be described in the following manner:

The authors confirm contribution to the paper as follows: ***study conception and design:*** *Divakhar A/L Rajah, Nur Ziadah Binti Harun;* ***data collection:*** *Divakhar A/L Rajah, Nur Ziadah Binti Harun;* ***analysis and interpretation of results:*** *Divakhar A/L Rajah, Nur Ziadah Binti Harun;* ***draft manuscript preparation:*** *Divakhar A/L Rajah, Nur Ziadah Binti Harun. All authors reviewed the results and approved the final version of the manuscript.*

An author name can appear multiple times, and each author name must appear at least once. For single authors, use the following wording:

The author confirms sole responsibility for the following: study conception and design, data collection, analysis and interpretation of results, and manuscript preparation.

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Appendix A: An Example

Authors including an appendix section should do so after the References section. Multiple appendices should all have headings in the style used above. They will automatically be ordered A, B, C etc.
Any extra data, equations, or information that is beneficial to the discussion of the paper should be included here. More appendices can be added as deemed necessary.